SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE:ITA0612 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 10%

QUESTIONS:50 **Total Marks:25**

Annexure A

Multiple Choice Questions (MCQ)

Code of Conduct:

I K. Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *K. Pujitha*

Annexure A

Short Answer Test

1. Bayes' Theorem is a fundamental concept in probability theory. Which of the following best describes its primary purpose in updating prior beliefs when new evidence is observed in a probabilistic model?
   a) To compute only prior probabilities
   b) To update prior beliefs using new evidence
   c) To eliminate irrelevant features
   d) To maximize classification accuracy

2. In the Naïve Bayes classifier, a strong assumption is made about the relationship between features. Which of the following statements correctly describes this assumption in practical machine learning applications?
   a) Features are completely dependent on each other
   b) Features are conditionally independent given the class
   c) Features must be continuous
   d) Features are always correlated

3. Maximum Likelihood Estimation is widely used for parameter estimation in probabilistic models. Which of the following options correctly explains the objective of MLE when fitting a model to observed data?
   a) Maximize prior probability
   b) Minimize hypothesis space
   c) Maximize likelihood of observed data
   d) Reduce computational complexity

4. The Minimum Description Length principle is often used in model selection. Which of the following best explains how MDL balances model complexity and goodness of fit in learning algorithms?
   a) Prefers highly complex models always
   b) Prefers simplest model that explains data well
   c) Ignores data noise completely
   d) Focuses only on training accuracy

5. The Bayes Optimal Classifier is considered theoretically ideal in classification tasks. Which of the following statements best describes how it achieves the minimum possible classification error?
   a) By selecting random hypotheses
   b) By averaging predictions over all hypotheses
   c) By using only prior probabilities
   d) By minimizing dataset size

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Bayes' Theorem

Correct Answer: b)

To update prior beliefs using new evidence.

Explanation: Bayes' Theorem is used to calculate the probability of an event based on previously available information and newly observed evidence. It updates the prior probability into a posterior probability.

Formula:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Example: A doctor uses a patient's symptoms and medical test results to update the probability of having a particular disease.

2. Naïve Bayes Classifier

Correct Answer: b)

Features are conditionally independent given the class.

Explanation: Naïve Bayes assumes that the features are independent of each other when the class label is known. This assumption makes probability calculations simple and faster.

Example: In spam detection, words such as "Free", "Offer" and "Prize" are treated as conditionally independent given the email class (Spam or Not Spam).

3. Maximum Likelihood Estimation (MLE)

Correct Answer: c)

Maximize likelihood of observed data.

Explanation: MLE is a statistical method used to estimate unknown parameters of a model. It selects parameter values that make the observed data most likely.

Example: MLE can estimate the mean and variance of a dataset by finding the parameter values that maximize the likelihood of the collected observations.

4. Minimum Description Length (MDL)

Correct Answer: b)

Prefers simplest model that explains data well.

Explanation: MDL is a model selection principle that balances model complexity and goodness of fit. It selects a model that explains the data effectively without unnecessary complexity.

Example: In machine learning, MDL helps avoid overfitting by preferring a simpler model that explains the training data well.

5. Bayes Optimal Classifier

Correct Answer: b)

By averaging predictions over all hypotheses.

Explanation: The Bayes Optimal Classifier considers all possible hypotheses and combines their predictions according to their posterior probabilities. It selects the class with the highest total weighted probability.

Example: In disease prediction, it considers different possible classification models and combines their predictions to obtain the most probable disease category.